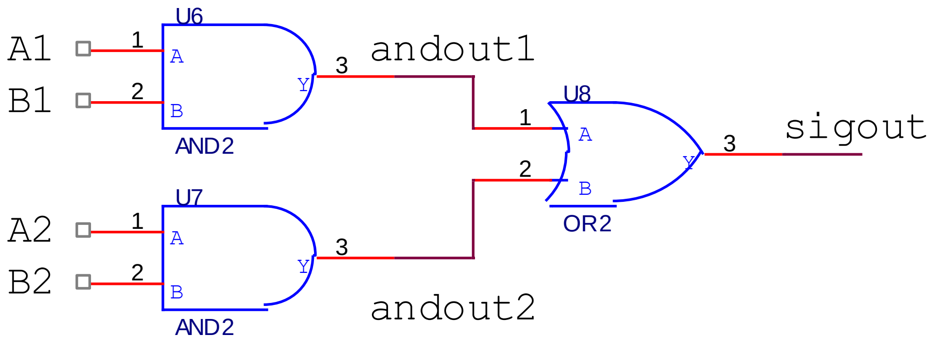
JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 1G: Module Review

Architecture Review




```
entity simplecircuit is
port (
   a1,b1,a2,b2 : in std_logic;
   sigout : out std_logic));
end simplecircuit;


architecture behavior of simplecircuit is
signal andout1, andout2 : std_logic;
begin

   -- U6 and U7 are component instantiations
   sigout <= andout1 or andout2; -- This is CSA
   U6: entity simpleAnd port map (a=>a1,b=>b1,y=>andout1);
   U7: entity simpleAnd port map (a=>a2,b=>b2,y=>andout2);
   -- All three assignments above happen concurrently (order irrelevant)

end behavior;
```

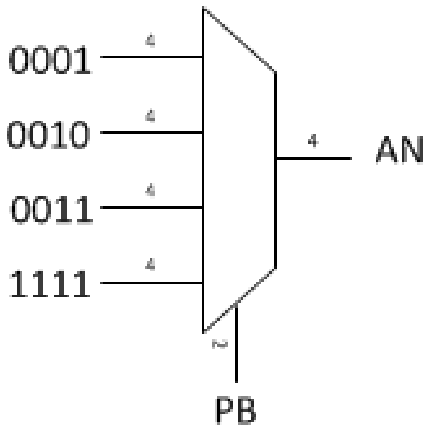
In this module, you…

- Recall how FPGAs operate in general
- Describe combinational logic using VHDL
- Compare different styles for describing combinational logic
- Construct a simple design using combinational logic only

Test your knowledge…

- Write the VHDL code for a conditional that describes the following table:

PB	AN
00	0001
01	0010
10	0011
11	1111



- Use both types of conditional statements covered so far
  - Refer back to the "with/select" statement in Module 1E, Slide 6 and "when/else" statement in Module 1E, Slide 3